



ZENTRA HELPDESK · INTEGRATION GUIDE

Video Conferencing Setup

A practical walkthrough for connecting Google Meet, Microsoft Teams, Zoom, and Cisco Webex to the Video Bridge integration. Includes credential checklists, exact navigation paths, and troubleshooting for every common error.

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1 · How the integration works

When an agent clicks **Start Bridge Call** on an incident, the Helpdesk server does this:

- 1 Reads the **Active provider** from `Settings → Integrations → Video Bridge`.
- 2 Loads the credentials saved for that provider.
- 3 Calls the provider's API to create a real meeting.
- 4 Stamps the join URL onto the incident and notifies the on-call team.

TIP Only **one** provider is active at a time. The system stores credentials for all four simultaneously, so you can switch providers without re-entering anything.

Provider implementations live in `server/src/lib/bridge/`. The **Test Connection** button creates and immediately deletes a real meeting — the only reliable way to verify credentials end-to-end.

2 · Where to enter credentials

- 1 Sign in to Helpdesk as an admin.
- 2 Open **Settings** → **Integrations**.
- 3 Scroll to **Video Bridge (Incident Bridge Calls)**.
- 4 Pick a provider from the **Active provider** dropdown.
- 5 Fill in the fields that appear, click **Save**.
- 6 Click **Test Connection**. ✓ *Connection successful* = done.

Screenshot slot

Settings → Integrations → Video Bridge card with the four provider options.

3 · Google Meet



Google Meet

OAuth 2.0 + long-lived Refresh Token · Setup ~15 min · Tokens never expire if consent screen is Internal or Published

What you need

Field	Source
OAuth Client ID	Google Cloud Console → Credentials → OAuth 2.0 Client IDs
OAuth Client Secret	Same credential entry
Refresh Token	Generated via OAuth Playground

3.1 Create a Google Cloud project

- 1 Open console.cloud.google.com.
- 2 Top-left **project picker** → **New Project**.
- 3 Name: `Zentra Video Meet`. If the **Organization** dropdown shows your Workspace domain, keep it selected.
- 4 **Create** and wait ~10 s.

Screenshot slot

Google Cloud Console — New Project dialog.

3.2 Enable the Calendar API

From the project, open **APIs & Services** → **Library**, search for **Google Calendar API**, click **Enable**. Optionally also enable **Google Meet API** for richer metadata.

NOTE Skipping this step is the #1 cause of `403 accessNotConfigured` when you click Test Connection.

3.3 Configure the OAuth consent screen

- 1 Open **APIs & Services** → **OAuth consent screen**.

- 2 User type: **Internal** (if on Workspace) or **External** (personal Gmail).
- 3 Fill in App name, support email, developer email → **Save and continue**.
- 4 Skip Scopes → continue.
- 5 External only: add the authorizing user under **Test users**.
- 6 If status is "Testing" and user type is External, click **Publish app** — this is what gives the refresh token an indefinite lifetime.

3.4 Create the OAuth client

- 1 **APIs & Services** → **Credentials** → + **Create Credentials** → **OAuth client ID**.
- 2 **Application type**: must be **Web application** (other types can't use the refresh-token flow).
- 3 Name: `Zentra Helpdesk Bridge`.
- 4 **Authorized JavaScript origins**:
`https://developers.google.com`
- 5 **Authorized redirect URIs**:
`https://developers.google.com/oauthplayground`
- 6 Click **Create**. Copy **Client ID** and **Client Secret** from the dialog — the secret is shown only at creation.

Screenshot slot

Create OAuth client — Web application form with JS origin and redirect URI filled.

3.5 Generate a refresh token via OAuth Playground

- 1 Open developers.google.com/oauthplayground.
- 2 Click the ⚙ **gear** (top right) → tick **Use your own OAuth credentials** → paste your Client ID and Secret → Close.
- 3 Step 1 — in the **Input your own scopes** box paste:

`https://www.googleapis.com/auth/calendar.events`

Click **Authorize APIs**, sign in, click **Allow**.

- 4 Step 2 — click **Exchange authorization code for tokens**. Copy the **Refresh token** (starts `1//...`).

Screenshot slot

OAuth Playground Step 2 showing the refresh token after exchange.

3.6 Paste into Helpdesk

Field	Value
Active provider	Google Meet
OAuth Client ID	from 3.4
OAuth Client Secret	from 3.4
Refresh Token	from 3.5

Common Google Meet errors

Error	Cause & fix
<code>accessNotConfigured</code> / 403	Calendar API not enabled → step 3.2.
<code>unauthorized_client</code>	Client Secret rotated, or app type isn't "Web application" → step 3.4.
<code>invalid_grant</code>	Refresh token revoked / cross-bound / has whitespace → regenerate in 3.5.
"Request access" page	Wrong Google account signed in → switch via the avatar top-right.

4 • Microsoft Teams



Microsoft Teams

Azure AD App Registration with application permissions · Setup ~30 min · Requires Global Admin consent + Teams policy grant

What you need

Field	Source
Tenant ID	Azure AD Overview
Client ID	App Registration Overview
Client Secret	App Registration → Certificates & secrets
Organizer User ID	UPN (email) or Object ID of the meeting organizer

4.1 Register the app

- 1 Open portal.azure.com as a tenant admin.
- 2 Search **Azure Active Directory** (now **Microsoft Entra ID**).
- 3 **App registrations** → **+ New registration**.
- 4 Name: .
- 5 Supported account types: **Single tenant**.
- 6 Redirect URI: leave blank.
- 7 **Register**.

Screenshot slot

Azure — App Registration form.

4.2 Copy Tenant ID and Client ID

On the Overview page: copy **Application (client) ID** and **Directory (tenant) ID**.

4.3 Create a Client Secret

- 1 **Certificates & secrets** → **+ New client secret**.

- 2 Expires: 24 months (or per policy).
- 3 **Add.**
- 4 **Copy the Value column immediately** — shown once only.

CRITICAL If you don't copy the secret value at this exact moment, delete the secret and create a new one. Azure will not show it again.

4.4 Grant API permissions

- 1 **API permissions** → **+ Add a permission** → **Microsoft Graph** → **Application permissions** (not Delegated).
- 2 Tick `OnlineMeetings.ReadWrite.All` → **Add permissions**.
- 3 Click **Grant admin consent for <tenant>** — needs Global Admin.

4.5 Authorize the app to impersonate the organizer

Teams meeting creation via application permissions also requires an **application access policy**:

```
Install-Module MicrosoftTeams -Scope CurrentUser
Connect-MicrosoftTeams
New-CsApplicationAccessPolicy -Identity Zentra-Bridge -AppIds "<APP_ID>" -Description
Grant-CsApplicationAccessPolicy -PolicyName Zentra-Bridge -Identity "<UPN>"
```

Wait ~30 min for the policy to propagate.

4.6 Paste into Helpdesk

Field	Value
Active provider	Microsoft Teams
Tenant ID	from 4.2
Client ID	from 4.2
Client Secret	from 4.3
Organizer User ID	UPN e.g. <code>support@yourcompany.com</code>

Common Teams errors

Error	Cause & fix
<code>AADSTS7000215: Invalid client secret</code>	Secret wrong or rotated → regenerate in 4.3.
<code>Forbidden</code> on meeting create	Application access policy not granted, or wrong UPN → step 4.5.
<code>AADSTS50011</code>	Admin consent not granted on Graph permissions → step 4.4.

5 · Zoom



Zoom

Server-to-Server OAuth · Setup ~5 min · No user consent, no expiry

What you need

Field	Source
Account ID	Zoom Marketplace → App Credentials
Client ID	Same page
Client Secret	Same page

5.1 Create the Server-to-Server OAuth app

- 1 Open marketplace.zoom.us, sign in.
- 2 Top-right **Develop** → **Build App**.
- 3 Pick **Server-to-Server OAuth** → **Create**.
- 4 Name: `Zentra Helpdesk Bridge`.

5.2 Copy credentials

On **App Credentials**: copy **Account ID**, **Client ID**, **Client Secret**.

5.3 Configure scopes

- 1 **Information**: fill mandatory fields.
- 2 **Scopes** → add:
 - `meeting:write:admin` — create meetings
 - `meeting:read:admin` — read meetings (Test Connection uses this)
- 3 **Continue** → **Activation** → **Activate your app**.

5.4 Paste into Helpdesk

Field	Value
Active provider	Zoom
Account ID	from 5.2
Client ID	from 5.2
Client Secret	from 5.2

Common Zoom errors

Error	Cause & fix
<code>Invalid client_id or client_secret</code>	Wrong creds → re-copy from 5.2.
Invalid access token immediately after save	App not activated → 5.3 → Activate.
<code>4711</code> missing scopes	Scopes not granted → add both, reactivate.

6 · Cisco Webex



Cisco Webex

Bot Token · Setup ~3 min · No OAuth, no expiry (bot tokens are permanent)

What you need

Field	Source
Bot Token	developer.webex.com → your bot
Site URL	Your Webex domain, e.g. <code>company.webex.com</code>

6.1 Create a bot

- 1 Open developer.webex.com/my-apps, sign in.
- 2 **Create a New App → Bot.**
- 3 Bot name: `Zentra Helpdesk`; bot username must be globally unique, e.g. `zentra-helpdesk@webex.bot`.
- 4 **Add Bot.**
- 5 Copy the **Bot Access Token** — shown once. Regenerate via My Apps → bot → Regenerate access token if lost.

PRODUCTION NOTE Use a **bot token** in production — permanent. **Personal Access Tokens** from developer.webex.com expire after 12 h and are fine only for short-lived testing.

6.2 Site URL

The domain you sign in to, e.g. `company.webex.com` — no `https://` and no path.

6.3 Paste into Helpdesk

Field	Value
Active provider	Cisco Webex
Bot Token	from 6.1
Site URL	from 6.2

Common Webex errors

Error	Cause & fix
<code>401 Unauthorized</code>	Wrong or expired token → regenerate.
<code>403 Forbidden – meeting:create</code>	Bot not licensed for meetings — use a licensed user's PAT or an Integration.
<code>404</code> on <code>/meetings</code>	Wrong site URL → no <code>https://</code> , no trailing path.

7 • Provider comparison

Provider	Auth method	Setup	Lifetime	Admin consent
Google Meet	OAuth 2.0 + Refresh Token	~15 min	Indefinite if Published / Internal	Workspace admin if Internal
Teams	OAuth 2.0 Client Credentials	~30 min	Secret 24 mo	Yes — Global Admin
Zoom	Server-to-Server OAuth	~5 min	Indefinite	No
Webex	Bot Token	~3 min	Indefinite	No

8 · Troubleshooting cheat sheet

1. `[none] No video bridge provider is configured` — Active provider is still set to None.
2. `Incomplete <provider> configuration` — one of the required fields is blank.
3. **Test passes but agents see no link** — check incident detail page; check server logs for the provider-specific error.
4. **Worked yesterday, fails today** — rotated Google secret, expired Webex PAT (12 h), or revoked OAuth consent. Bot tokens, refreshed Google tokens (Published), and Zoom S2S do not expire.

PRO TIP When Test Connection fails, server logs print the exact provider response. Open `server` stdout (or your hosting log viewer) and look for lines starting with `[googlemeet]`, `[teams]`, `[zoom]`, `[webex]`. The error is always more specific in the log than in the toast.